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1. (a) Given: $(1-x)^2 \frac{d^2 y}{dx^2} + xy = e^{x+3}$

$$\frac{d^2 y}{dx^2} + \frac{x}{(1-x)^2} y = \frac{e^{x+3}}{(1-x)^2}$$

$$P(x) = 0 \quad \theta(x) = \frac{x}{(1-x)^2} \quad \text{and} \quad \phi(x) = \frac{e^{x+3}}{(1-x)^2}$$

at $x=0$.

$$P(0) = 0 \quad \theta(0) = 0 \quad \text{and} \quad \phi(0) = e^3$$

hence the point $x=0$ is ordinary point.

(b). Given: $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + x^2 y = 0$

$$\frac{d^2 y}{dx^2} - \frac{x}{(1-x^2)} \frac{dy}{dx} + \frac{x^2}{(1-x^2)} y = 0$$

$$P(x) = -\frac{x}{(1-x^2)} \quad \theta(x) = \frac{x^2}{(1-x^2)} \quad \text{and} \quad \phi(x) = 0$$

at $x=1$

$$P(1) = -\infty \quad \theta(1) = \infty \quad \text{and} \quad \phi(1) = 0$$

hence the point $x=1$ is ordinary singular point.

2. Given: $8x^2 \frac{d^2 y}{dx^2} + 10x \frac{dy}{dx} + (x-1)y = 0$ ——— 1.

Step 1 Test

$$\frac{d^2 y}{dx^2} + \frac{10x}{8x^2} \frac{dy}{dx} + \frac{(x-1)}{8x^2} y = 0$$

$$P(x) = \frac{10x}{8x^2} \quad \theta(x) = \frac{(x-1)}{8x^2} \quad \text{and} \quad \phi(x) = 0$$

at $x=0$, $P(0) = \infty$ $\theta(0) = \infty$ and $\phi(0) = 0$.

hence the point $x=0$ is regular singular point.

2. Step 2: let $y(x) = \sum_{k=0}^{\infty} a_k x^{\lambda+k}$ ————— 2

$$y'(x) = \sum_{k=0}^{\infty} (\lambda+k) a_k x^{\lambda+k-1}$$
 ————— 3

$$y''(x) = \sum_{k=0}^{\infty} (\lambda+k)(\lambda+k-1) a_k x^{\lambda+k-2}$$
 ————— 4

Step 3: substitute equations 2, 3 and 4 into 1.

$$8x^2 \sum_{k=0}^{\infty} (\lambda+k)(\lambda+k-1) a_k x^{\lambda+k-2} + 10x \sum_{k=0}^{\infty} (\lambda+k) a_k x^{\lambda+k-1} + (x-1) \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0$$

$$\sum_{k=0}^{\infty} 8(\lambda+k)(\lambda+k-1) a_k x^{\lambda+k} + \sum_{k=0}^{\infty} 10(\lambda+k) a_k x^{\lambda+k+1} + \sum_{k=0}^{\infty} a_k x^{\lambda+k+1} - \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0$$

Step 4: Put $k=0$ to get indicial equation.

$$\sum_{k=0}^{\infty} 8\lambda(\lambda-1) a_0 x^{\lambda} + \sum_{k=0}^{\infty} 10\lambda a_0 x^{\lambda+1} + \sum_{k=0}^{\infty} a_0 x^{\lambda+1} - \sum_{k=0}^{\infty} a_0 x^{\lambda} = 0$$

$$(8\lambda(\lambda-1) + 10\lambda - 1) a_0 x^{\lambda} = 0$$

$$8\lambda^2 - 8\lambda + 10\lambda - 1 = 0$$

$$8\lambda^2 + 2\lambda - 1 = 0$$

$$\lambda_1 = \frac{1}{4} \quad \text{and} \quad \lambda_2 = -\frac{1}{2}$$

Step 5: let $k=k-1$ for third term.

$$\sum_{k=0}^{\infty} 8(\lambda+k)(\lambda+k-1) a_k x^{\lambda+k} + \sum_{k=0}^{\infty} 10(\lambda+k) a_k x^{\lambda+k+1} + \sum_{k=0}^{\infty} a_{k-1} x^{\lambda+k} - \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0$$

$$8(\lambda+k)(\lambda+k-1) a_k + 10(\lambda+k) a_k + a_{k-1} - a_k = 0$$

$$[8(\lambda+k)(\lambda+k-1) + 10(\lambda+k) - 1] a_k = a_{k-1}$$

$$a_k = \frac{a_{k-1}}{8(\lambda+k)(\lambda+k-1) + 10(\lambda+k) - 1}$$

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$$\text{for } k=1, \quad q_1 = \frac{-q_0}{8\lambda(\lambda) + 10\lambda - 1} = \frac{-q_0}{8\lambda^2 + 10\lambda - 1}$$

$$k=2, \quad q_2 = \frac{-q_1}{8(\lambda+2)}$$

$$2. \text{ for } k=1, \quad q_1 = \frac{-q_0}{8(\lambda+1)\lambda + 10(\lambda+1) - 1} = \frac{-q_0}{(\lambda+1)[8\lambda+10] - 1}$$

$$k=2, \quad q_2 = \frac{-q_1}{8(\lambda+2)(\lambda+1) + 10(\lambda+2) - 1} = \frac{-q_1}{(\lambda+2)[8(\lambda+1)+10] - 1}$$

$$q_2 = \frac{q_0}{[(\lambda+2)[8(\lambda+1)+10] - 1][(\lambda+1)[8\lambda+10] - 1]}$$

for, $\lambda_1 = 1/4$

$$q_1 = \frac{-q_0}{5/4(12) - 1} = \frac{-q_0}{14}$$

$$q_2 = \frac{-q_1}{9/4(18) - 1} = \frac{-2q_1}{79} = \frac{-2}{79} \times \left(\frac{-q_0}{14} \right)$$

$$q_2 = \frac{q_0}{553}$$

$$y_1(x) = q_0 + q_1 x + q_2 x^2 + \dots$$

$$y_1(x) = q_0 - \frac{q_0}{14} x + \frac{q_0}{553} x^2 + \dots$$

$$y_1(x) = q_0 \left(1 - \frac{1}{14} x + \frac{1}{553} x^2 + \dots \right)$$

for, $\lambda_2 = -1/2$

$$q_1 = \frac{-q_0}{2}$$

$$q_2 = \frac{-q_1}{20} = \frac{-1}{20} \left(\frac{-q_0}{2} \right) = \frac{q_0}{40}$$

2. Assume $y_2(x) = a_0 + a_1 x + a_2 x^2 + \dots$

$$y_2(x) = a_0 - \frac{a_0}{2}x + \frac{a_0}{40}x^2 + \dots$$

$$y_2(x) = a_0 \left(1 - \frac{1}{2}x + \frac{1}{40}x^2 + \dots \right)$$

Then,

$$y(x) = e_{\lambda_1}(y)_{\lambda_1} + e_{\lambda_2}(y)_{\lambda_2} \quad (1)$$

$$y(x) = a_0 \left(1 - \frac{1}{2}x + \frac{1}{40}x^2 + \dots \right) + a_0 \left(1 - \frac{1}{2}x + \frac{1}{40}x^2 + \dots \right)$$

3. Given $\frac{d^2y}{dx^2} + xy = x + 2$ ————— 1

Step 01. Test

$$p(x) = 0 \quad \theta(x) = x \quad \phi(x) = x + 2$$

at $x=0$, $p(0)=0$ $\theta(0)=0$ $\phi(0)=2$

Hence the point is ordinary point at $(x=0)$.

Step 02. let

$$y(x) = \sum_{n=0}^{\infty} a_n x^n \quad \text{————— 2}$$

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{————— 3}$$

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \quad \text{————— 4}$$

Step 03. Substitute eqns 2, 3 and 4 into 1.

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + x \sum_{n=0}^{\infty} a_n x^n = x + 2$$

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$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^{n+1} = x+2.$$

Step 4: let $n = k+2$ and $n = k-1$

$$\sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k + \sum_{k=1}^{\infty} a_{k-1} x^k = x+2.$$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=1}^{\infty} a_{n-1} x^n = x+2.$$

$$2a_2 x + \sum_{n=1}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=1}^{\infty} a_{n-1} x^n = x+2.$$

$$\frac{2a_2}{2} = \frac{2}{2} \quad a_2 = 1.$$

$$[(n+2)(n+1) a_{n+2} + a_{n-1}] x^n = 0.$$

$$a_{n+2} = \frac{-a_{n-1}}{(n+2)(n+1)}$$

for $n=1$

$$a_3 = \frac{-a_0}{2}$$

$n=2$

$$a_4 = \frac{-a_1}{12}$$

$n=3$

$$a_5 = \frac{-a_2}{20}$$

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$y(x) = a_0 + a_1 x + x^2 - \frac{a_0}{2} x^3 - \frac{a_1}{12} x^4 - \frac{a_2}{20} x^5 + \dots$$

$$y(x) = \left(1 - \frac{1}{2}x^3 + \dots\right) a_0 + \left(x - \frac{x^4}{12} + \dots\right) a_1 +$$